15. A wire of length L and cross-sectional area A is shown.

The resistance R of the wire is given by the relationship

$$R = \frac{\rho L}{A}$$

where ρ is the resistivity of the wire in $\Omega\,\text{m}$.

(a) The resistivity of aluminium is $2{\cdot}8 \times 10^{-8}\,\Omega\,\text{m}$.

Calculate the resistance of an aluminium wire of length $0{\cdot}82\,\text{m}$ and cross-sectional area $4{\cdot}0 \times 10^{-6}\,\text{m}^2$. **2**

Space for working and answer

15. **(continued)** **MARKS** DO NOT WRITE IN THIS MARGIN

(b) A student carries out an investigation to determine the resistivity of a cylindrical metal wire of cross-sectional area $4{\cdot}52 \times 10^{-6}\,m^2$.

The student varies the length L of the wire and measures the corresponding resistance R of the wire.

The results are shown in the table.

Length of wire L (m)	*Resistance of wire R* ($\times 10^{-3}\,\Omega$)
1·5	5·6
2·0	7·5
2·5	9·4
3·0	11·2
3·5	13·2

(i) Using the square-ruled paper on *Page 36*, draw a graph of R against L. **3**

(ii) Calculate the gradient of your graph. **2**

Space for working and answer

(iii) Determine the resistivity of the metal wire. **3**

Space for working and answer

[END OF QUESTION PAPER]

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